

Chapter 06 Momentum and Collisions

Momentum

The **linear momentum** $\vec{p}$ of a mass m moving with velocity $\vec{v}$ is defined:

$$\vec{p} = m\vec{v}$$

And the direction of the momentum is the same as velocity's.

The **time rate** of change of momentum of an object is equal to the **net force**.

$$\vec{F}_{net} = \frac{\Delta \vec{p}}{\Delta t}$$

And we can get

$$\vec{F}_{net} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) = m \frac{d\vec{v}}{dt} = m\vec{a}$$

so the relations are **equivalent expressions of Newton's second law**.

Impulse

The impulse $\vec{I}$ is defined as (for a single, constant force):

$$\vec{I} = \vec{F}\Delta t$$

The direction is the same as the force's

For the change of momentum, the impulse $\vec{J}$ has:

$$\vec{J} = \int_{t_i}^{t_f} \vec{F}(t) dt$$

and since $\vec{F}_{net} = \frac{d\vec{p}}{dt}$, we have

$$\Delta \vec{p} = \vec{J}$$

Impulse-Momentum Theorem

The impulse acting on an object equals to the change in momentum

$$\vec{F}\Delta t = \Delta \vec{p} = m\vec{v}_f - m\vec{v}_i$$

If the force is not constant, use the **average force**, which satisfies:

$$\vec{F}_{avg}\Delta t = \Delta \vec{p}$$

Momentum Conservation

Momentum is conserved for the system of objects

$$m_1\vec{v}_{1i} + m_2\vec{v}_{2i} = m_1\vec{v}_{1f} + m_2\vec{v}_{2f}$$

Collisions

For collisions, momentum is always conserved.

- Inelastic collisions: Kinetic energy is not conserved. Perfectly inelastic collisions lose kinetic energy most, occurring when the objects stick together.
- Elastic collisions: Kinetic energy is conserved.

Glancing Collisions

The conservation of momentum implies that the total momentum of the system in each direction:

$$\begin{aligned}m_1v_{1ix} + m_2v_{2ix} &= m_1v_{1fx} + m_2v_{2fx} \\m_1v_{1iy} + m_2v_{2iy} &= m_1v_{1fy} + m_2v_{2fy}\end{aligned}$$

Rocket Propulsion

An inelastic collision. Kinetic energy is increased. It has

$$Rv_{rel} = Ma$$

where

$$R = -\frac{dM}{dt}$$

, the rate at which the rocket loses mass. v_{rel} is the relative speed, $v_{rel} = v + dv - u$

And we can get the change of the velocity of a rocket:

$$v_f - v_i = v_{rel} \ln \frac{M_i}{M_f}$$

obtained by integrating

$$dv = -v_{rel} \frac{dM}{M}$$

, equivalent to the first equation.